LAB 02

Hands-On Git and Version Control

Name	Sidra-Tul-Muntaha
Registration Number	23-BSCS-084
Course Code	CS-649
Section	A
GitHub Username	sidra2244

Repository: DevOps
Submission Directory: Labs/Lab02

Figure Evidence — Screenshot Insertion Pages

Figure 1: Git installation verification

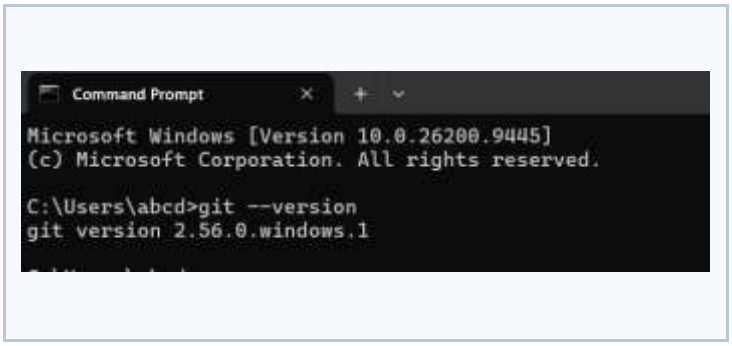


Figure 1. Git installation verification

Description: Shows Git installed and the terminal output confirming the installation.

Figure 2: Private GitHub repository

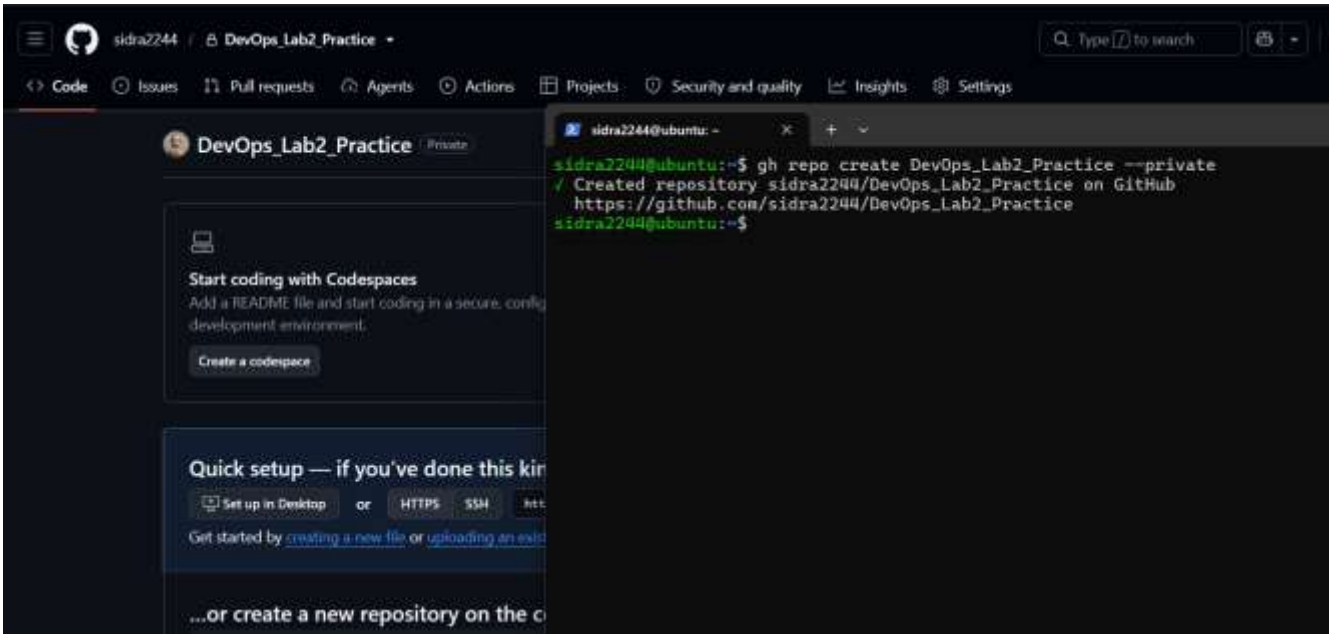


Figure 2. Private GitHub repository

Description: Shows DevOps_Lab2_Practice marked Private, alongside the required terminal prompt.

Figure 3: SSH key generation

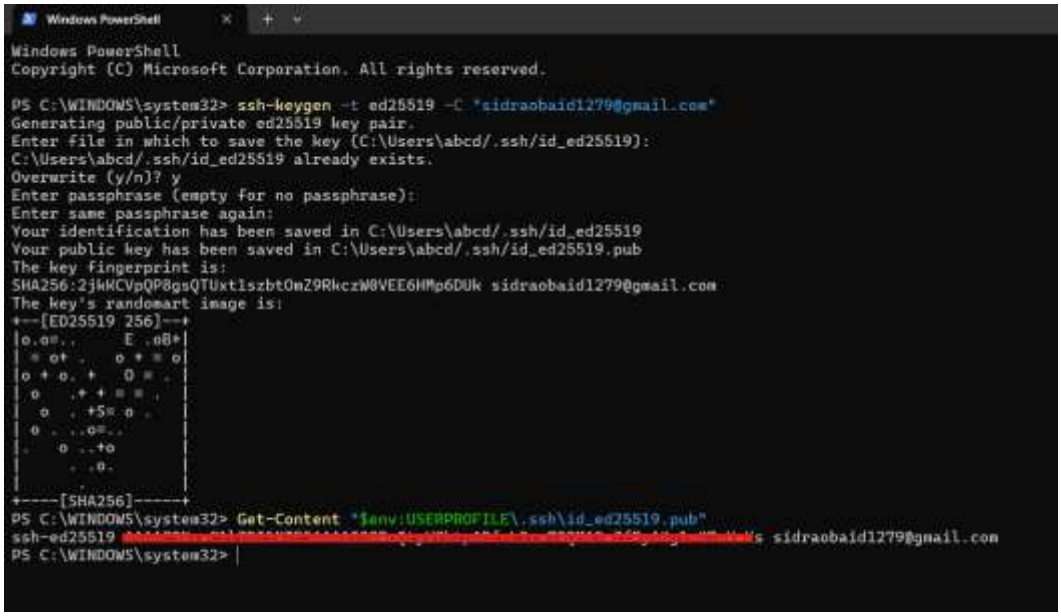


Figure 3. SSH key generation

Description: Shows the SSH key-generation command and result. Do not show private key contents.

Figure 4: SSH public key added to GitHub

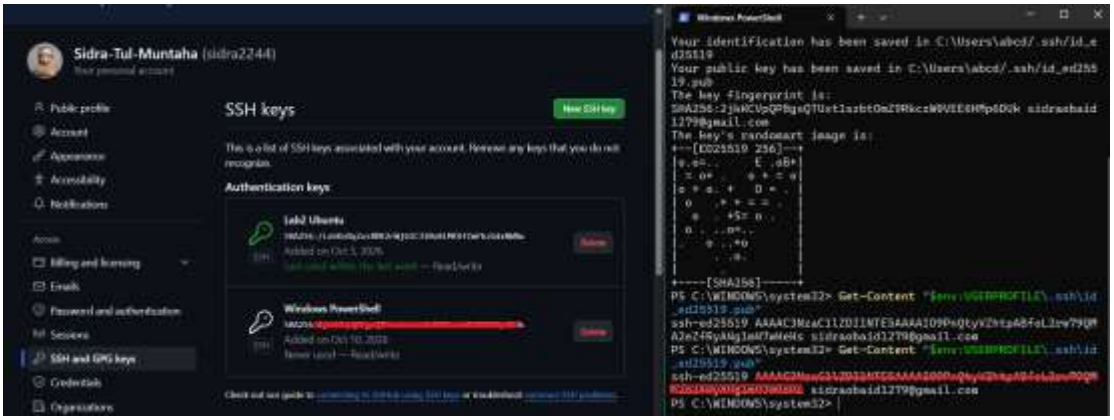


Figure 4. SSH public key added to GitHub

Description: Shows the public key entry under GitHub Settings → SSH and GPG keys.

Figure 5: Repository cloned using SSH

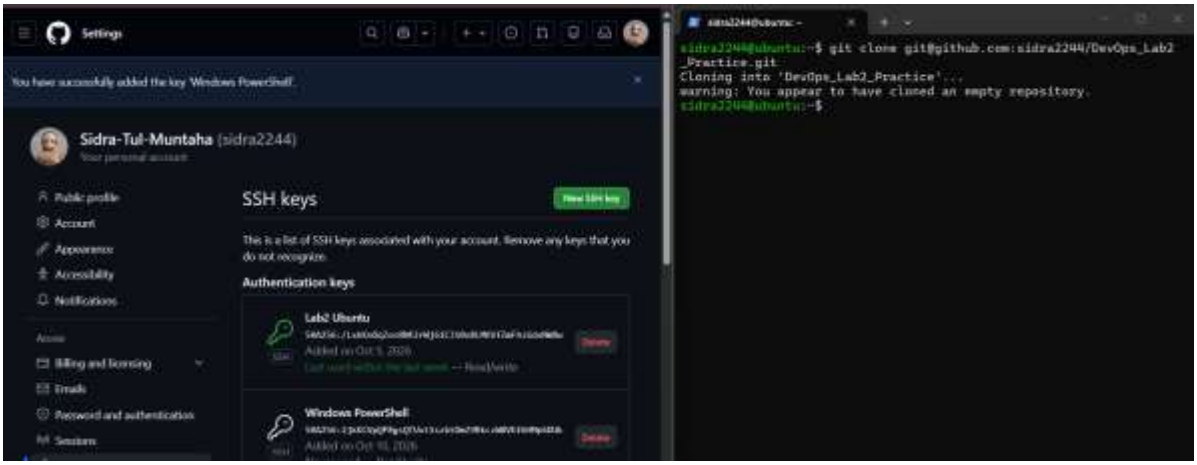


Figure 5. Repository cloned using SSH

Description: Shows the SSH clone command and successful result.

Figure 6: Git identity configuration

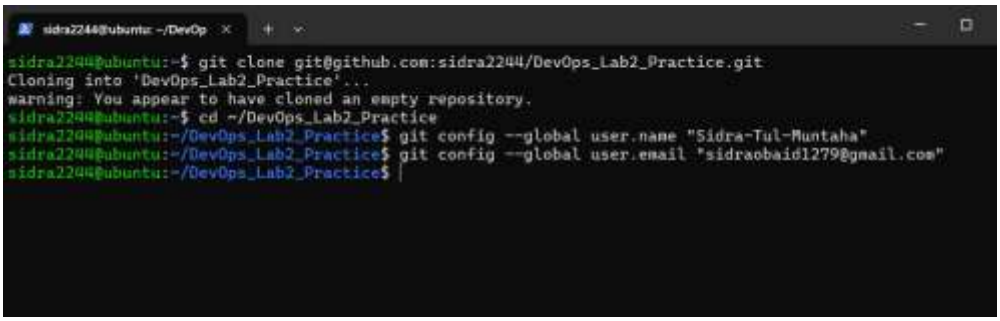


Figure 6. Git identity configuration

Description: Shows commands setting the Git username and email.

Figure 7: Git configuration verification

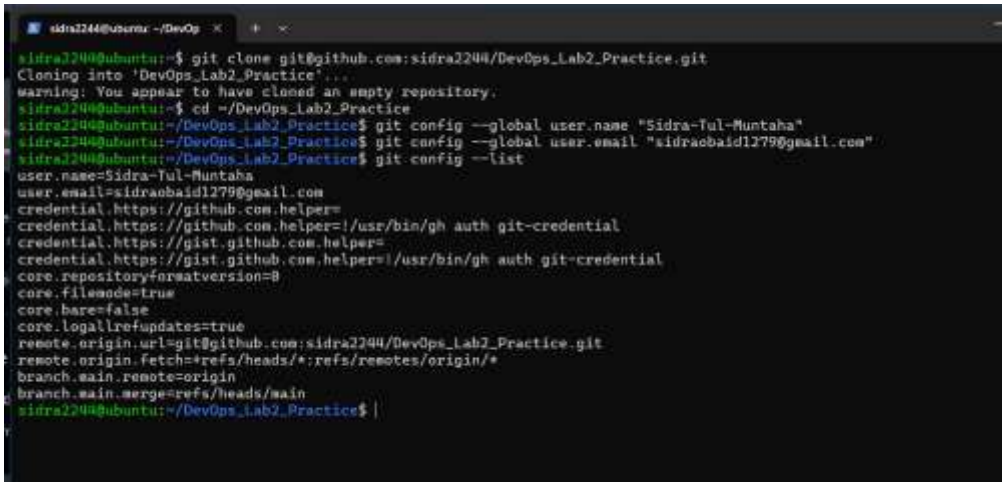


Figure 7. Git configuration verification

Description: Shows the output of git config --list.

Figure 8: Contents of the .git directory



Figure 8. Contents of the .git directory

Description: Shows ls -a .git and repository metadata.

Figure 9: Removing the .git directory



Figure 9. Removing the .git directory

Description: Shows removal of .git and the resulting git status error.

Figure 10: Reinitializing the repository

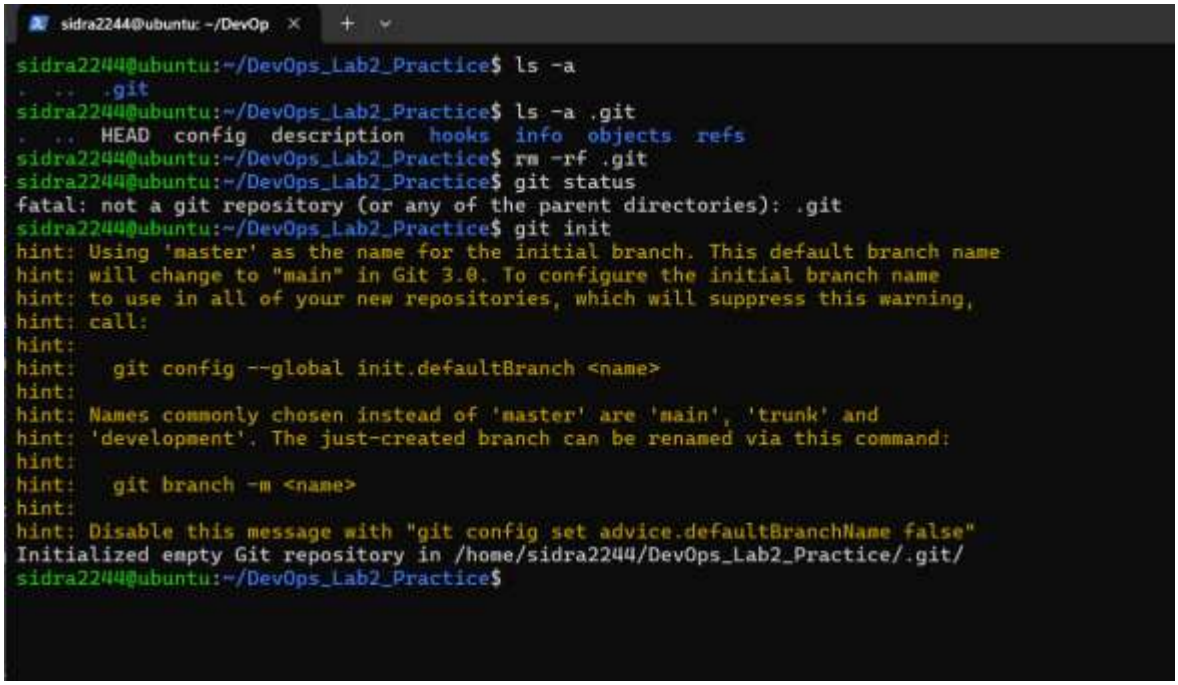


Figure 10. Reinitializing the repository

Description: Shows git init and its success message.

Figure 11: First commit

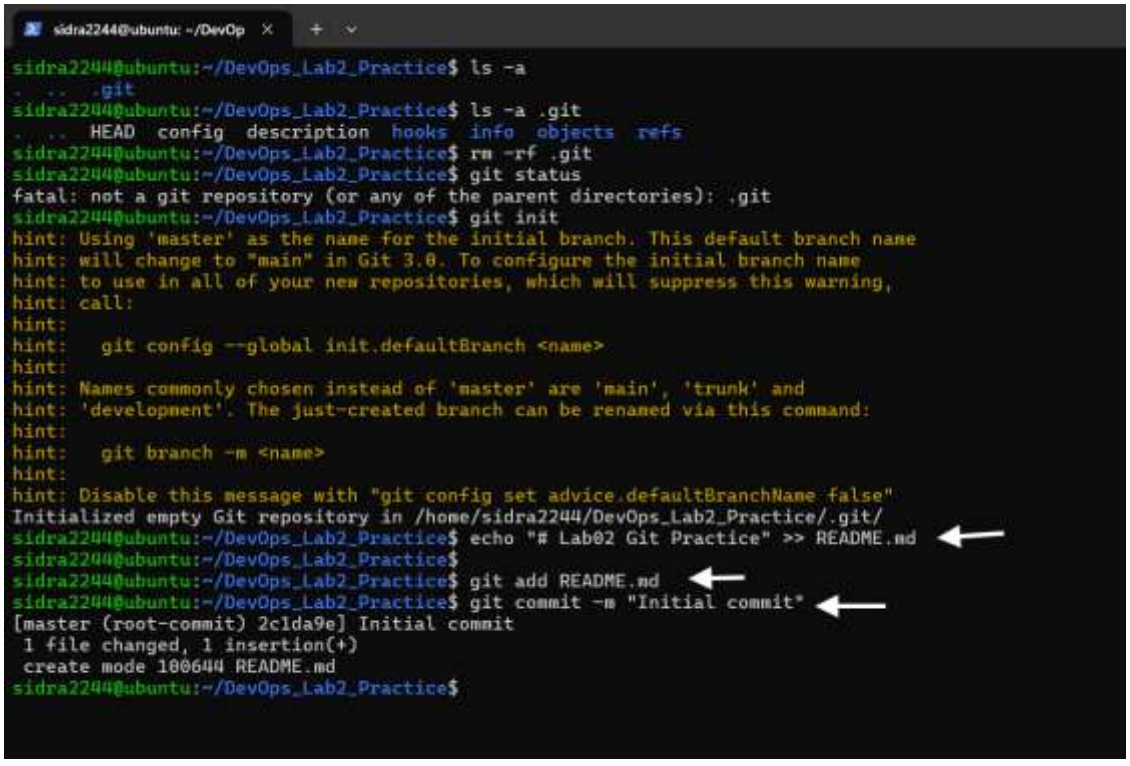


Figure 11. First commit

Description: Shows README.md staged and committed.

Figure 12: Pushing the repository to GitHub

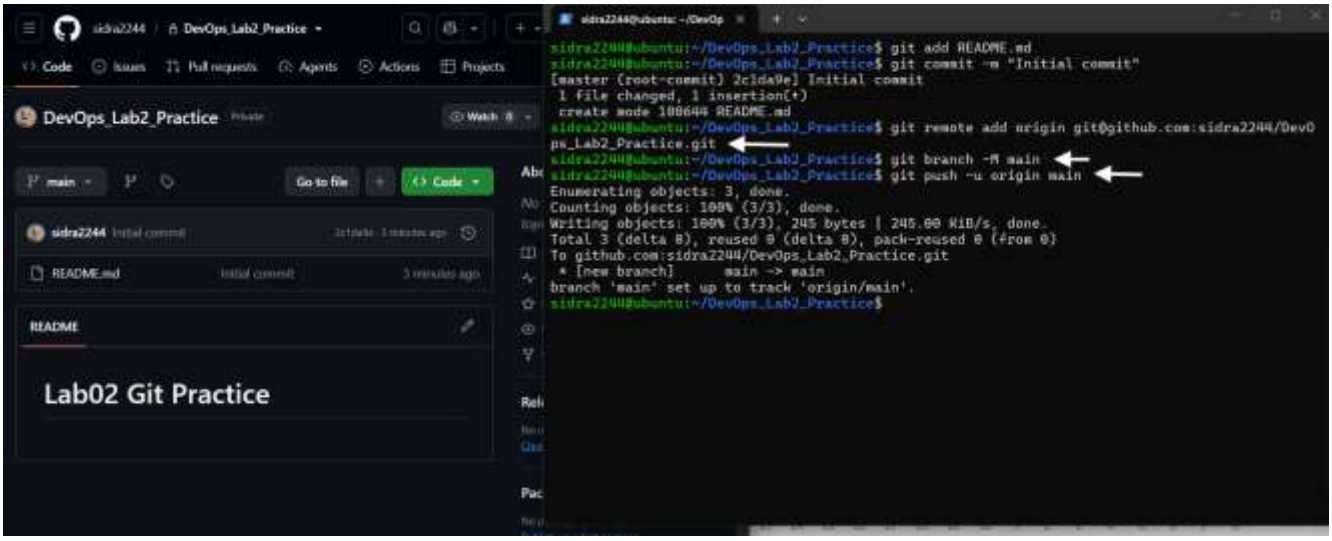


Figure 12. Pushing the repository to GitHub

Description: Shows the remote setup and successful push of main.

Figure 13: Checking file status

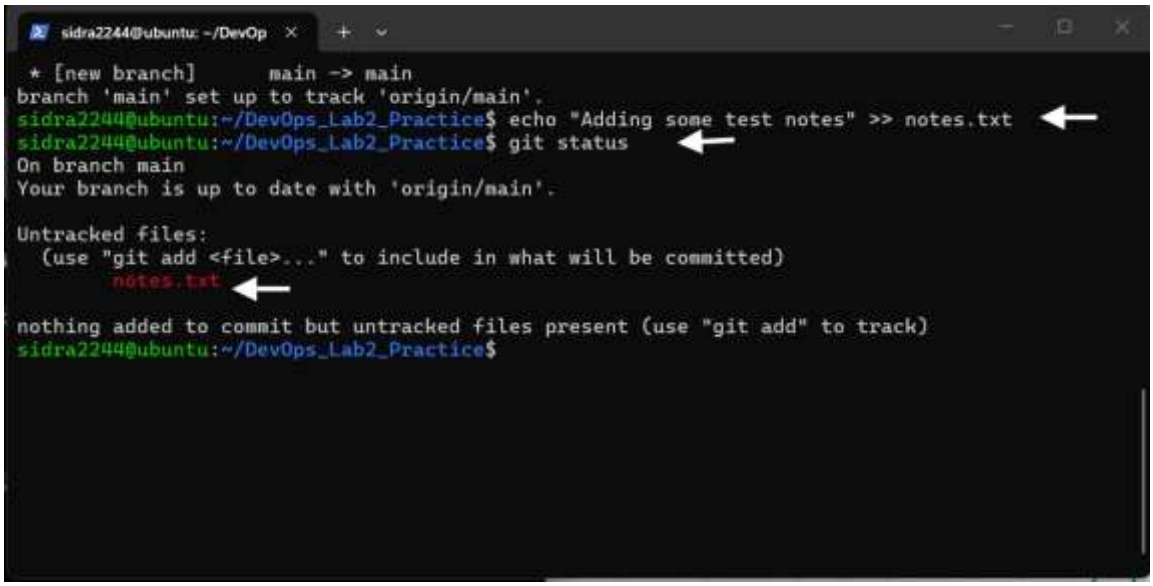


Figure 13. Checking file status

Description: Shows notes.txt as an untracked file after git status.

Figure 14: Staging and committing notes.txt

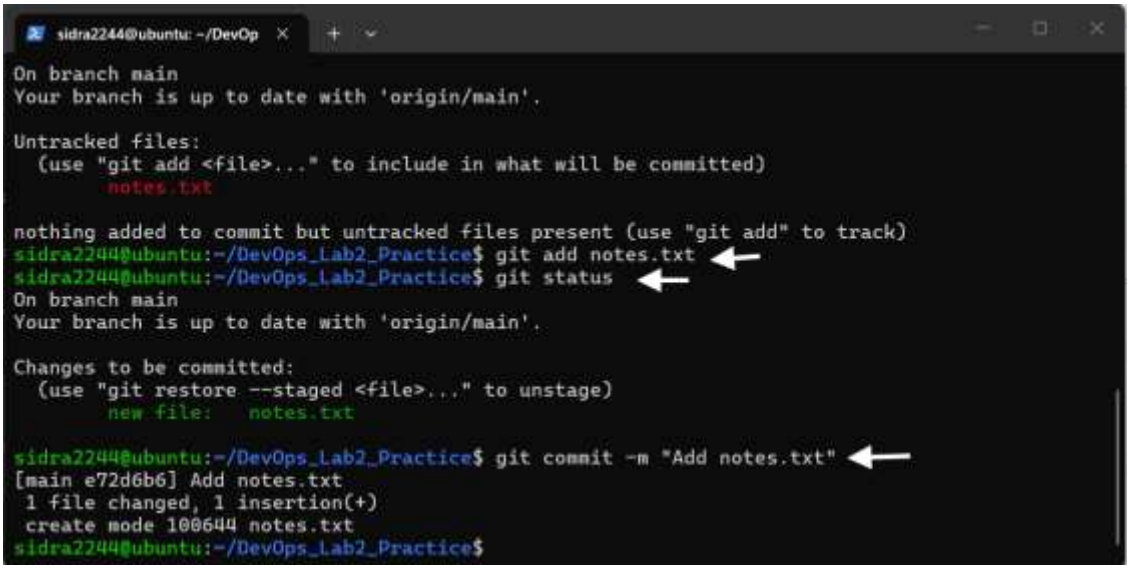


Figure 14. Staging and committing notes.txt

Description: Shows notes.txt staged and committed.

Figure 15: Creating a branch in GitHub GUI

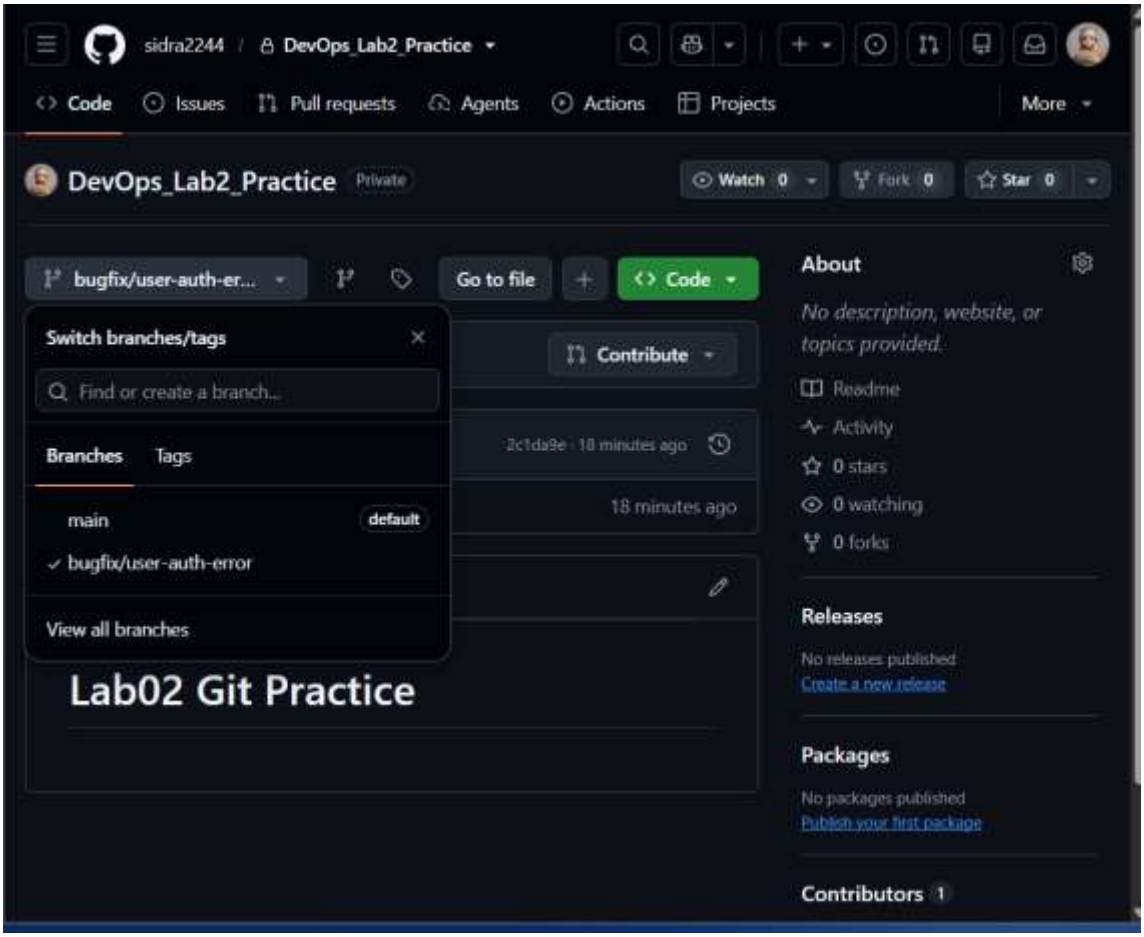


Figure 15. Creating a branch in GitHub GUI

Description: Shows the bugfix/user-auth-error branch created on GitHub.

Figure 16: Local bugfix branch

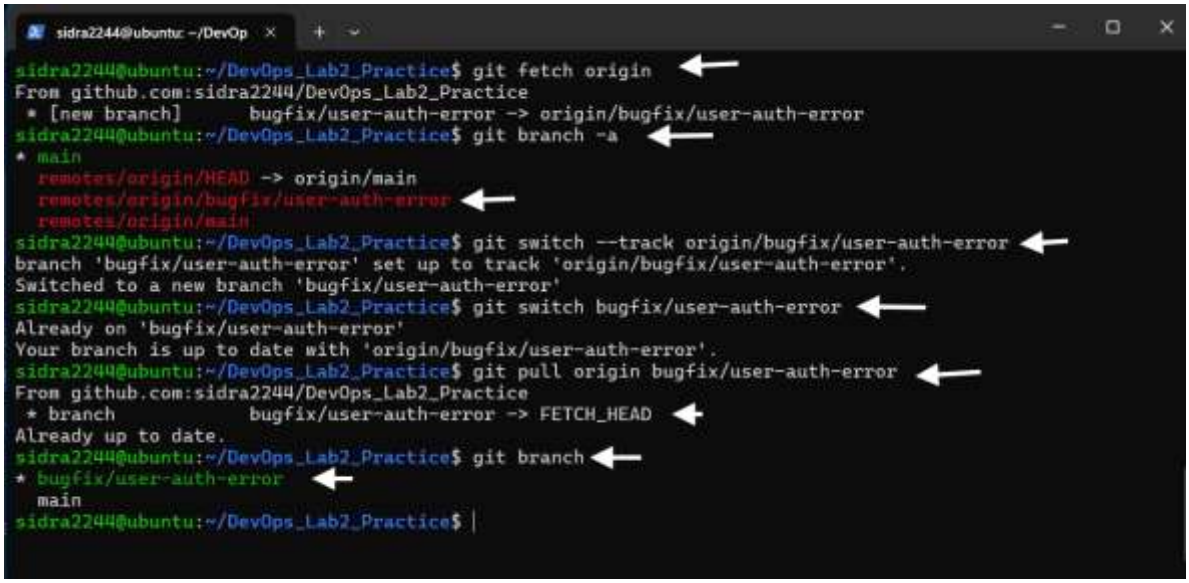


Figure 16. Local bugfix branch

Description: Shows the local branch list after synchronizing the branch.

Figure 17: Creating and pushing feature/db-connection

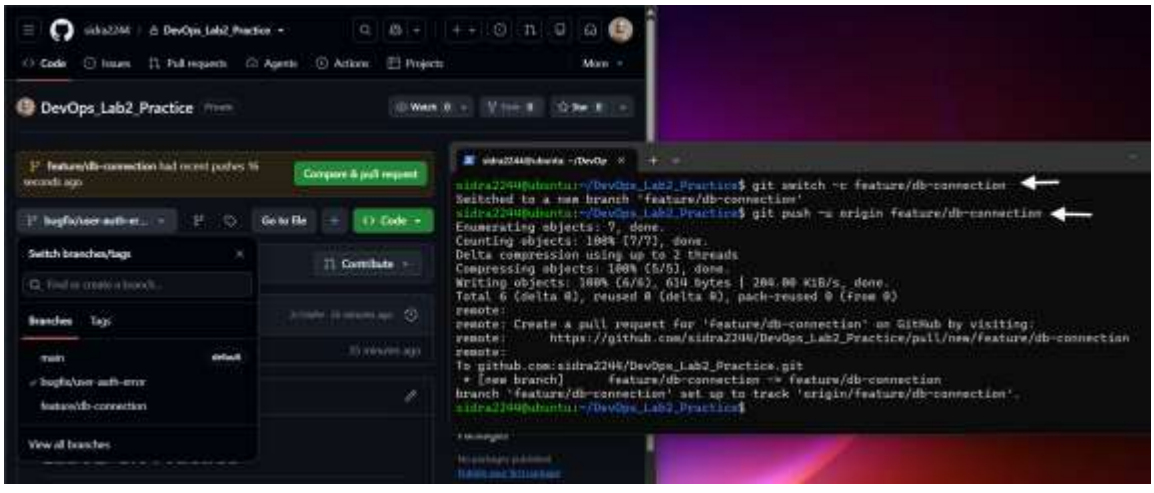


Figure 17. Creating and pushing feature/db-connection

Description: Shows the branch creation command and remote push.

Figure 18: Creating feature-1

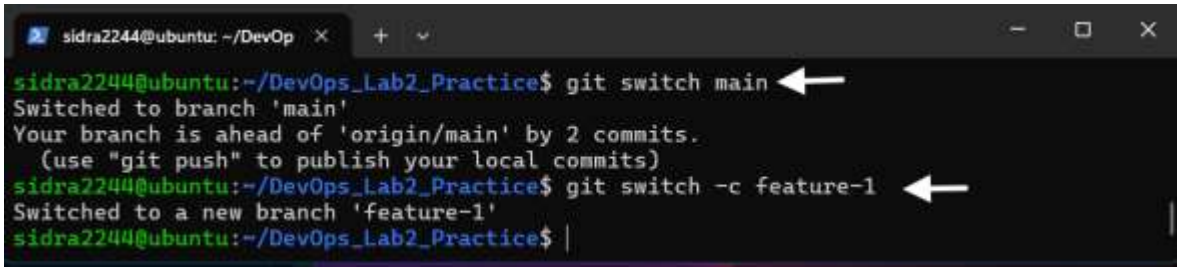


Figure 18. Creating feature-1

Description: Shows the command that creates and switches to feature-1.

Figure 19: Committing a feature change

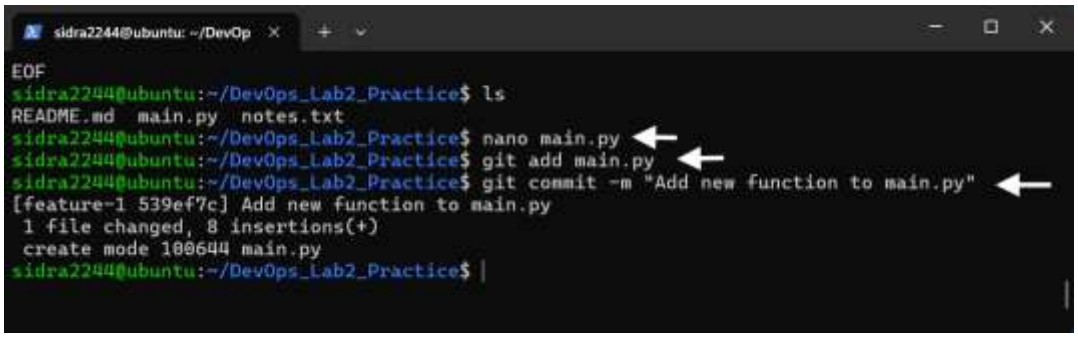


Figure 19. Committing a feature change

Description: Shows main.py staged and committed with the new function.

Figure 20: Merging feature-1 into main

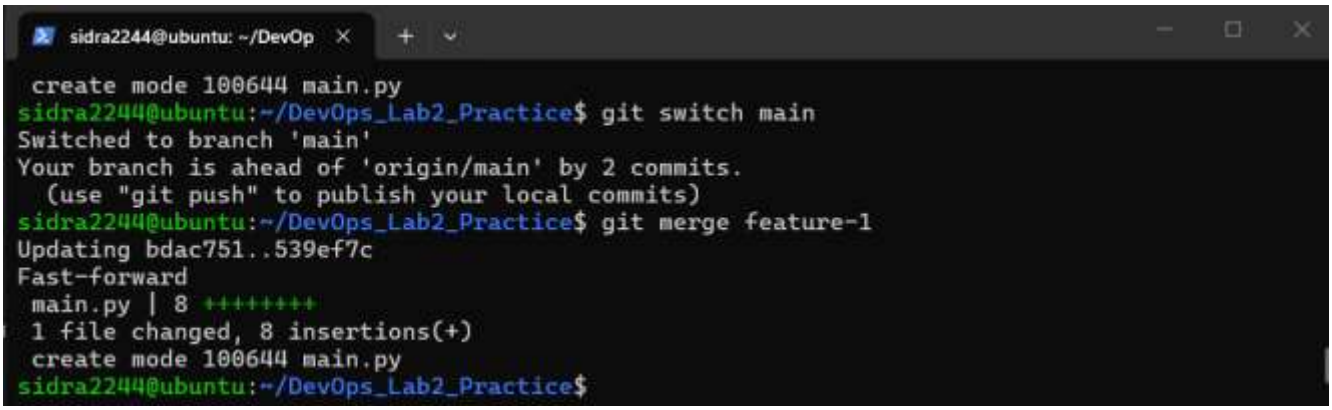


Figure 20. Merging feature-1 into main

Description: Shows switching to main and the merge result.

Figure 21: Pushing branches to GitHub

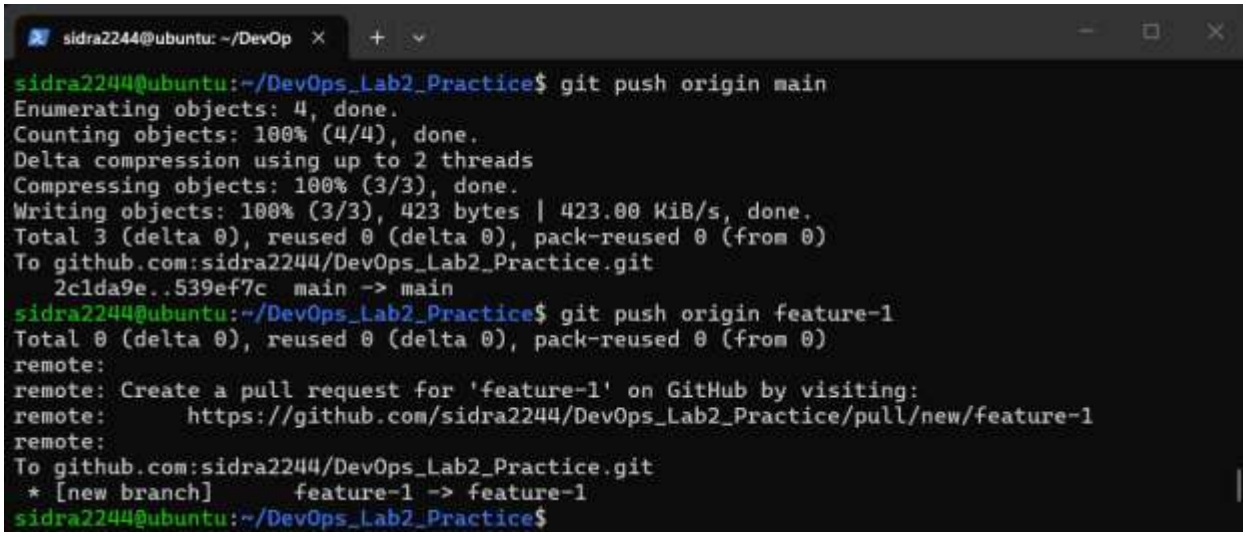


Figure 21. Pushing branches to GitHub

Description: Shows successful pushes of main and feature-1.

Figure 22: Pull request creation

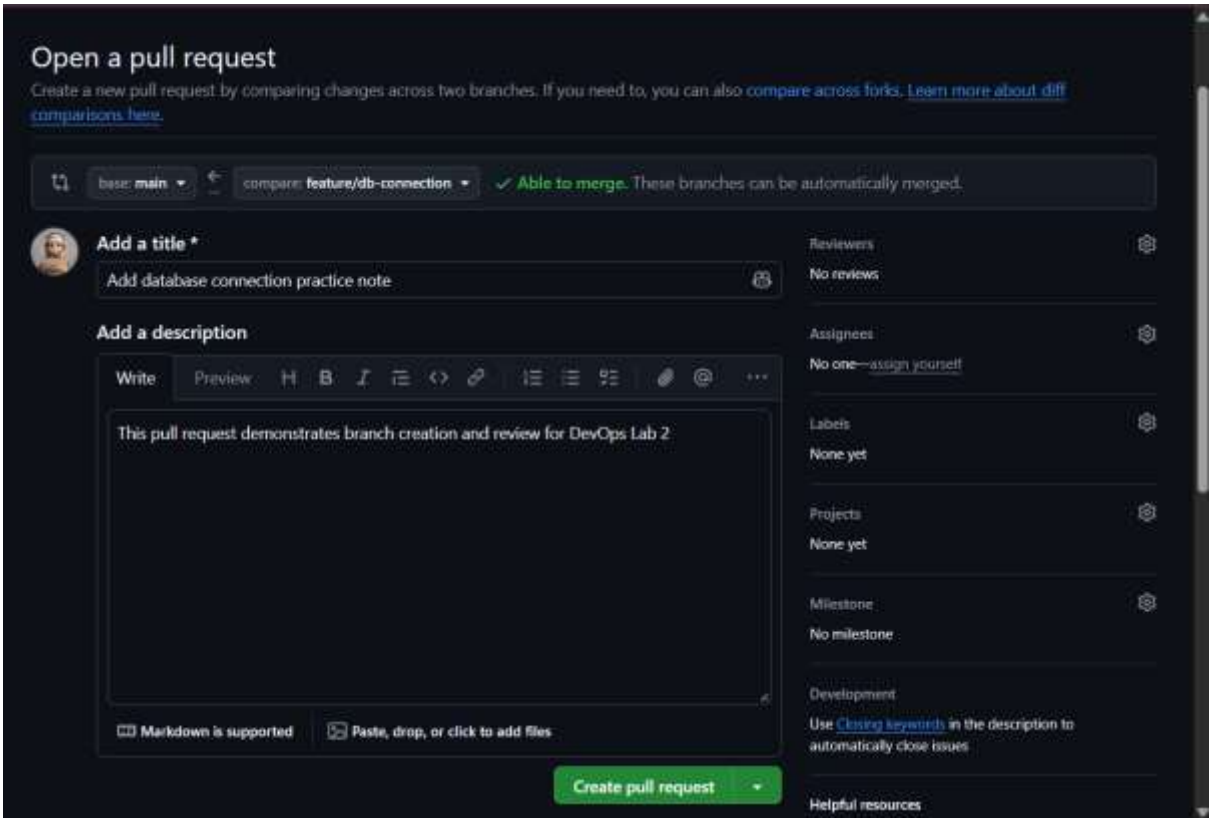


Figure 22. Pull request creation

Description: Shows a pull request from feature/db-connection into main.

Figure 23: Reviewing and merging a pull request

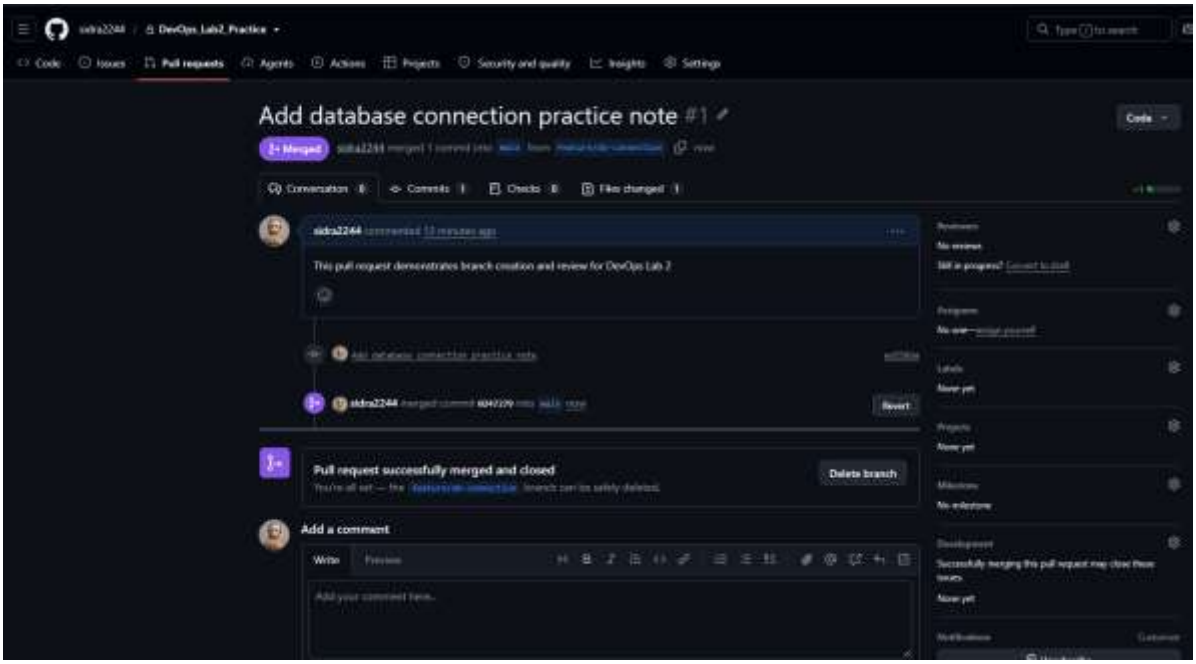


Figure 23. Reviewing and merging a pull request

Description: Shows the PR review and merge action in GitHub.

Figure 24: Deleting a merged branch

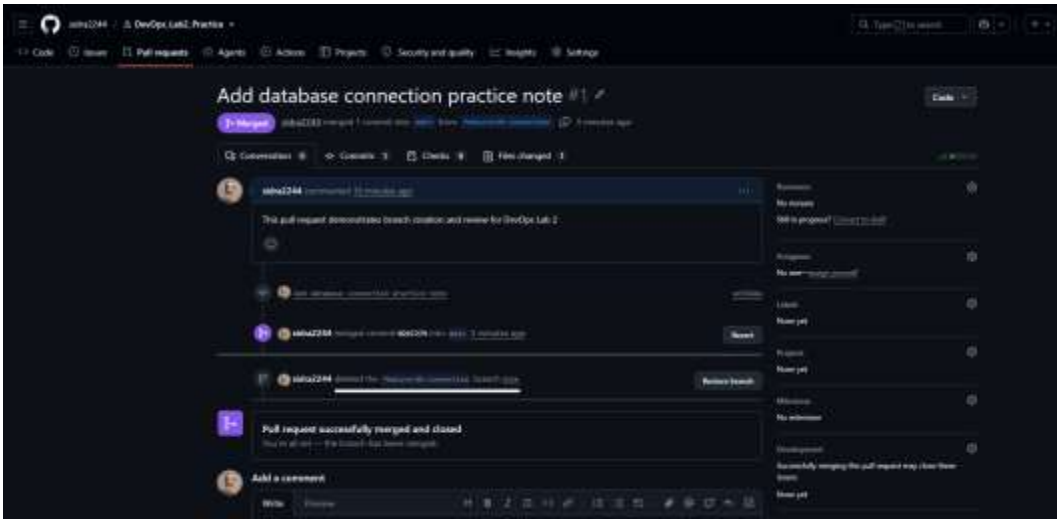


Figure 24. Deleting a merged branch

Description: Shows the feature branch deleted through GitHub.

Figure 25: Branch strategy overview

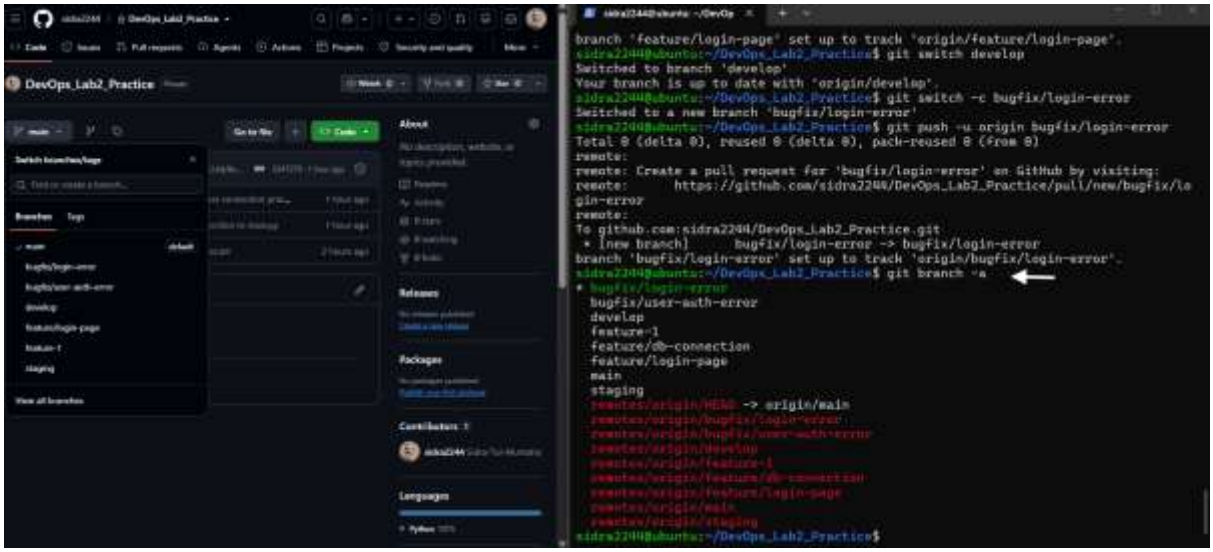


Figure 25. Branch strategy overview

Description: Shows develop, staging, feature/*, and bugfix/* branches.

Figure 26: Merges into develop and staging

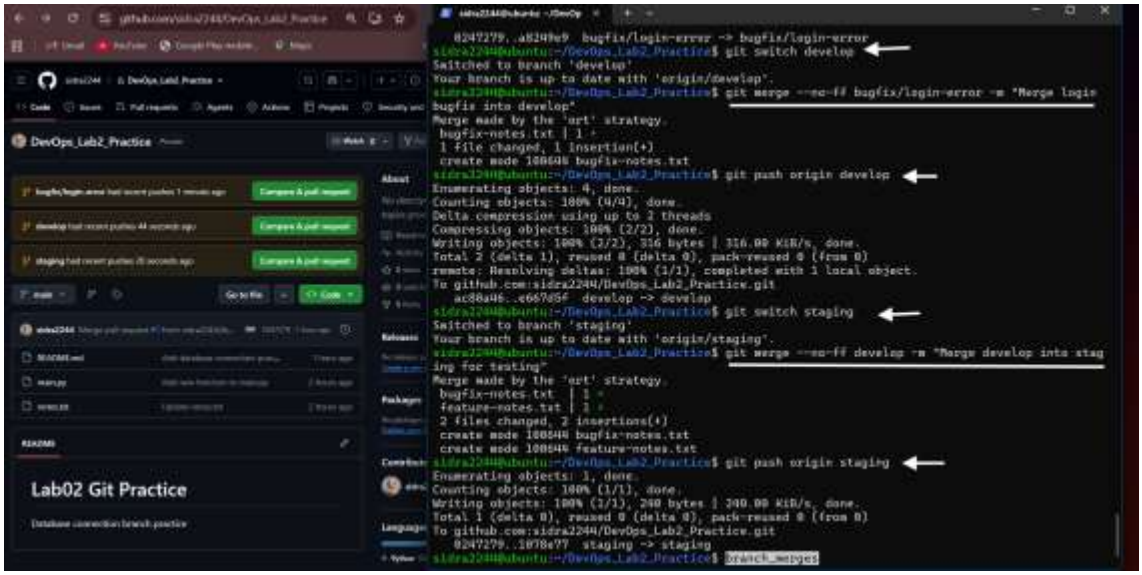


Figure 26. Merges into develop and staging

Description: Shows feature/bugfix work merged into develop and then staging.

Figure 27: Final merge into main

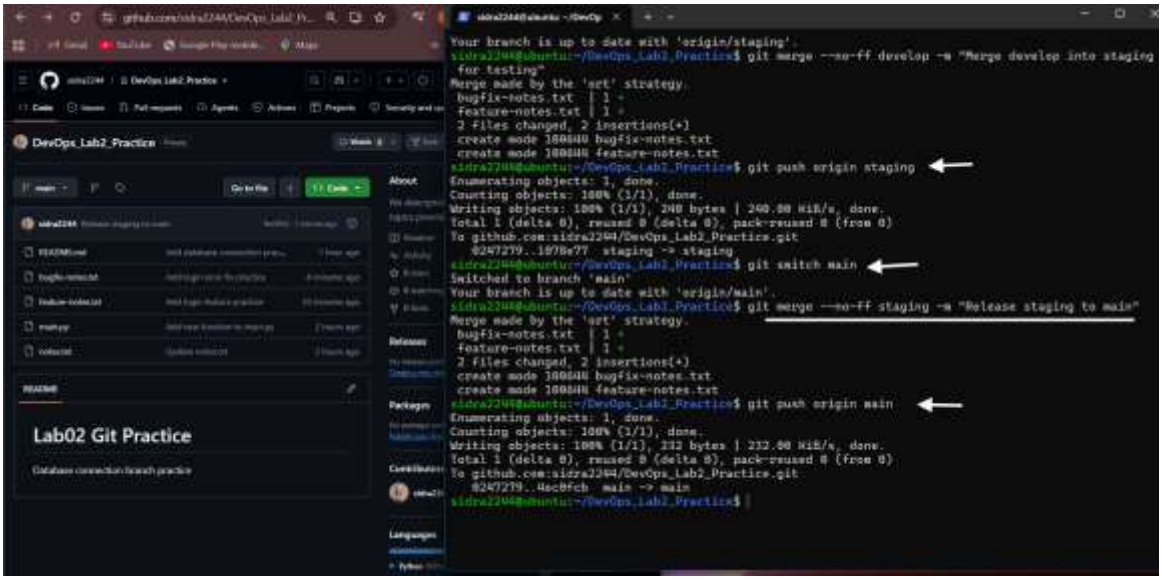


Figure 27. Final merge into main

Description: Shows staging merged into main.

Figure 28: Pull request title and description

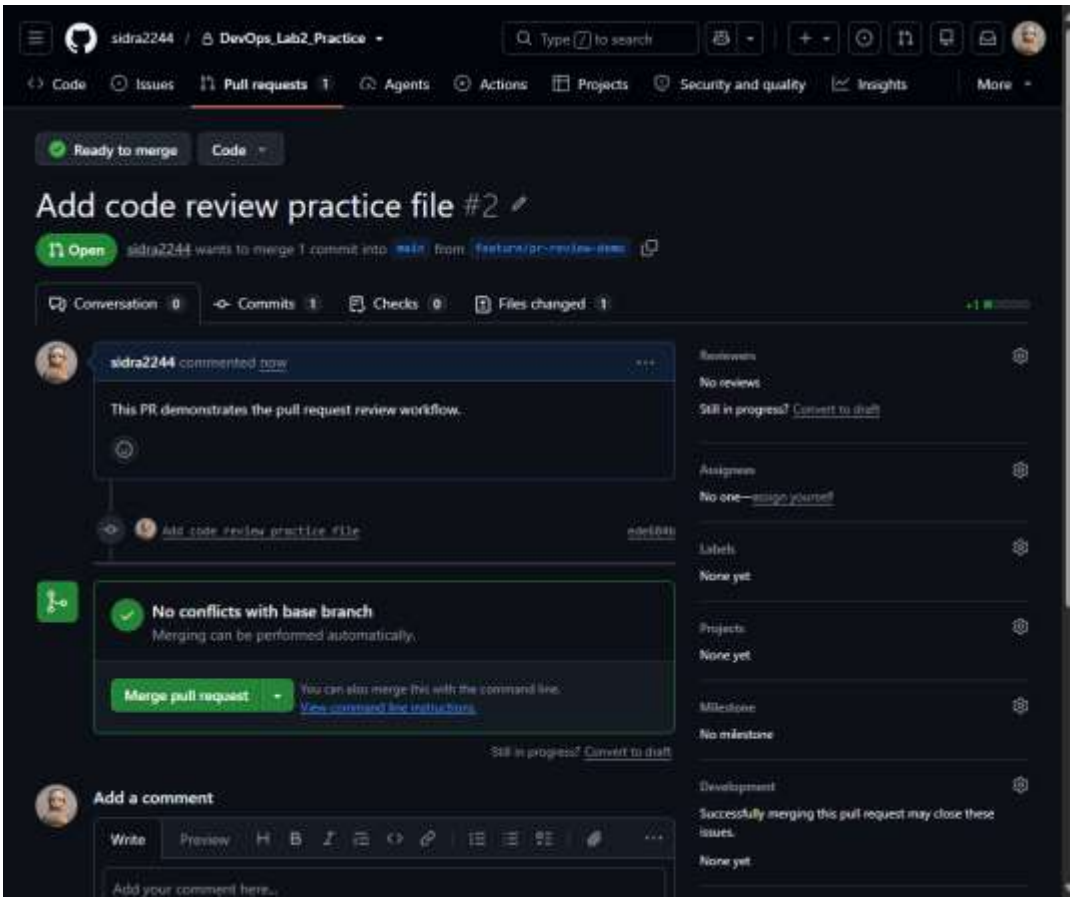


Figure 28. Pull request title and description

Description: Shows a feature-to-main PR with a clear title and description.

Figure 29: Reviewer assigned

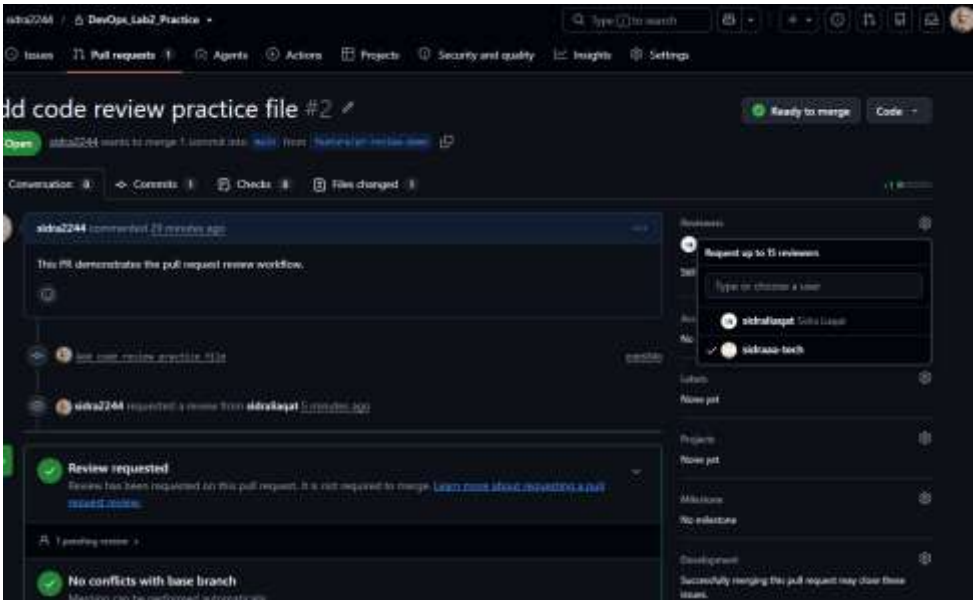


Figure 29. Reviewer assigned

Description: Shows a reviewer assigned to the pull request.

Figure 30: Reviewer requests changes

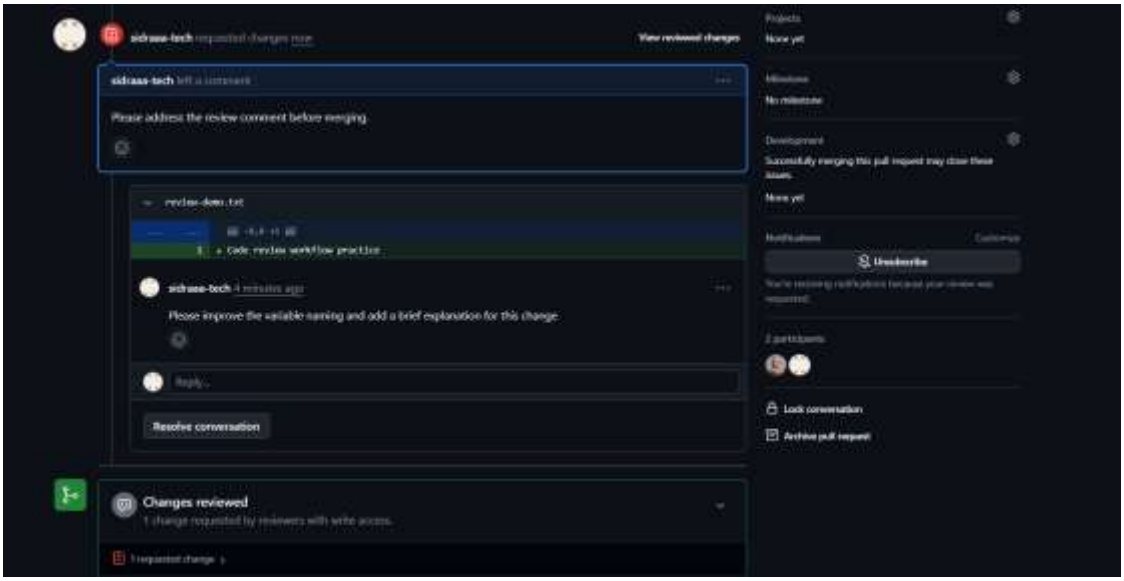


Figure 30. Reviewer requests changes

Description: Shows review comments and the request-changes state.

Figure 31: Practice pull request rejected or closed

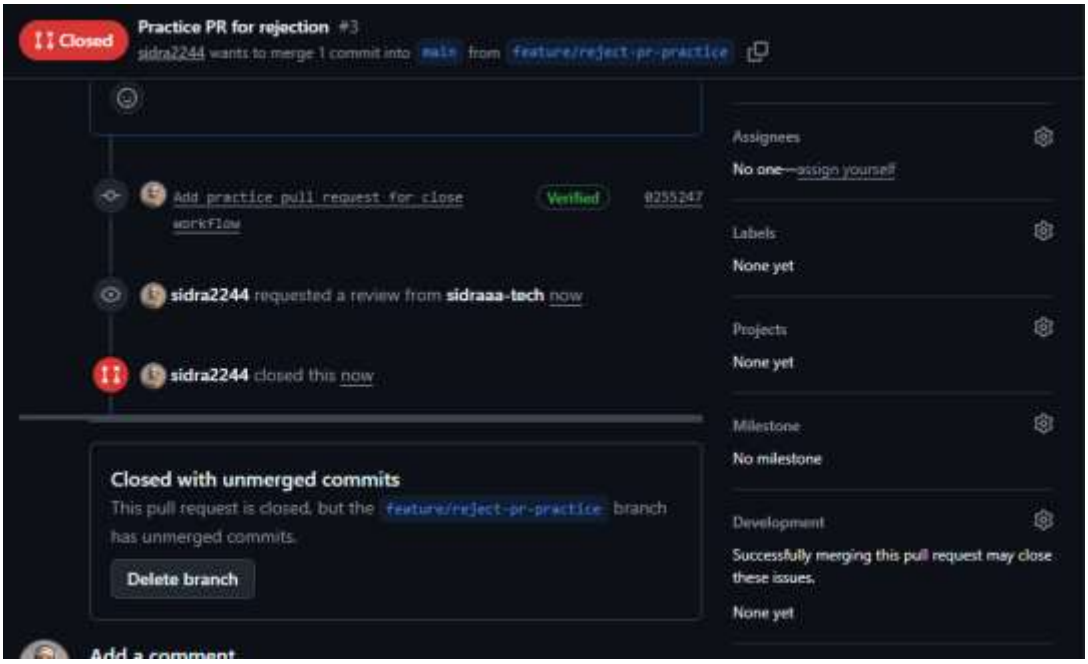


Figure 31. Practice pull request rejected or closed

Description: Shows a separate practice PR closed or rejected.

Figure 32: Pull request updated with commits

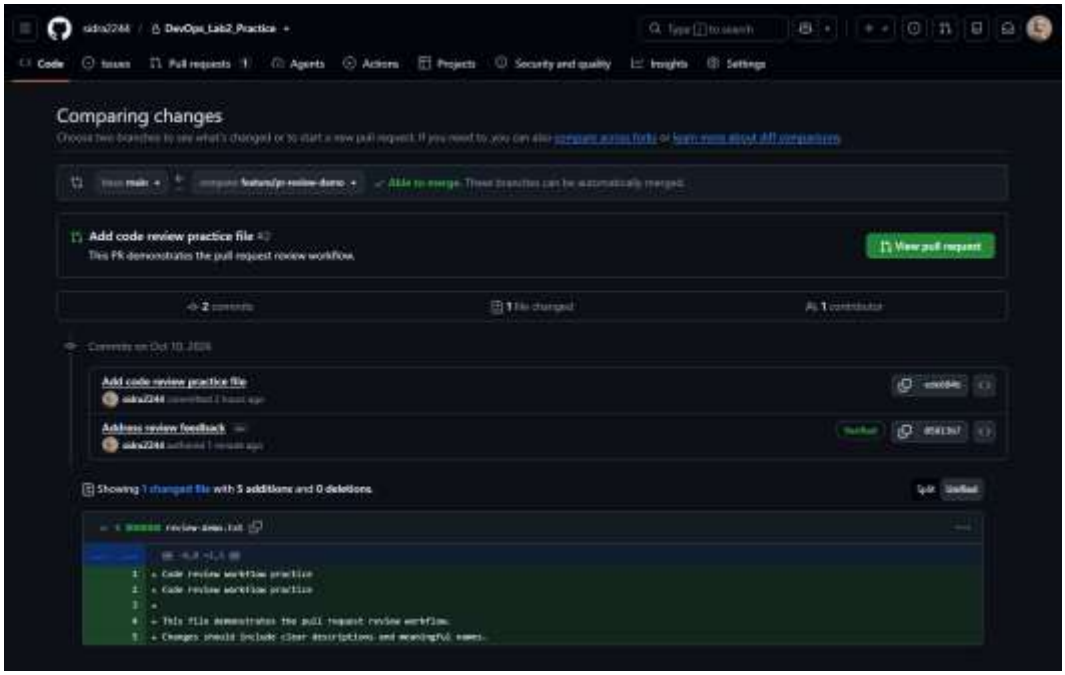


Figure 32. Pull request updated with commits

Description: Shows fixes pushed to the same branch and reflected in the PR.

Figure 33: Pull request merge confirmation

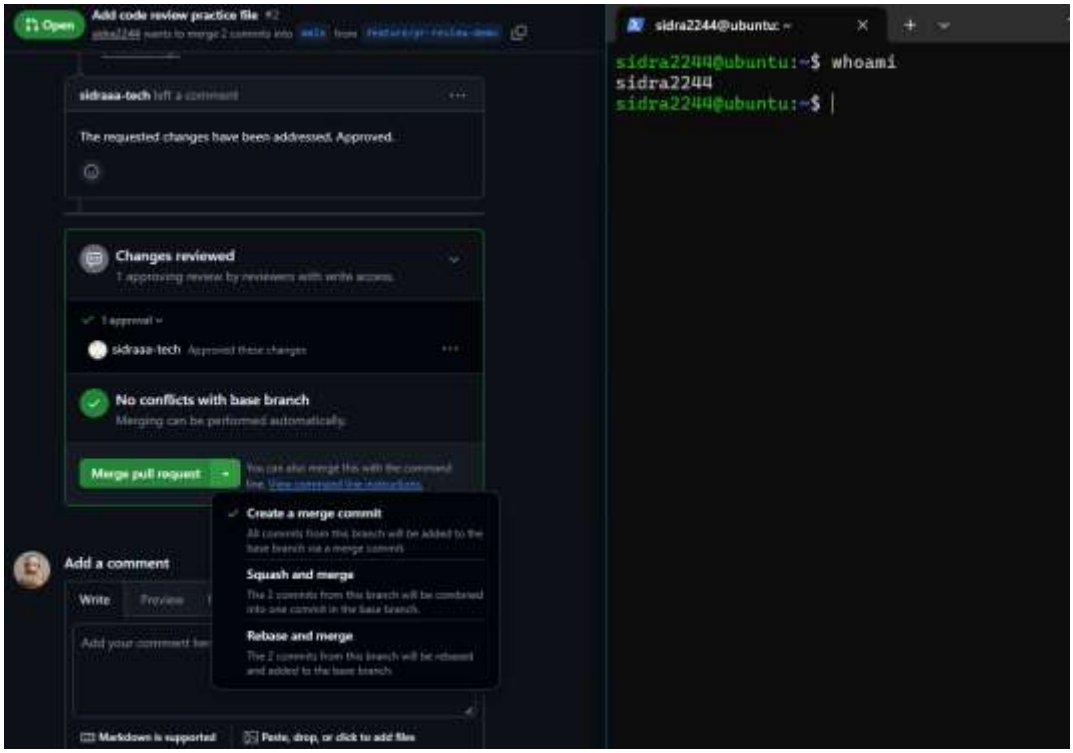


Figure 33. Pull request merge confirmation

Description: Shows the merge confirmation or selected merge method.

Figure 34: Pull request successfully merged

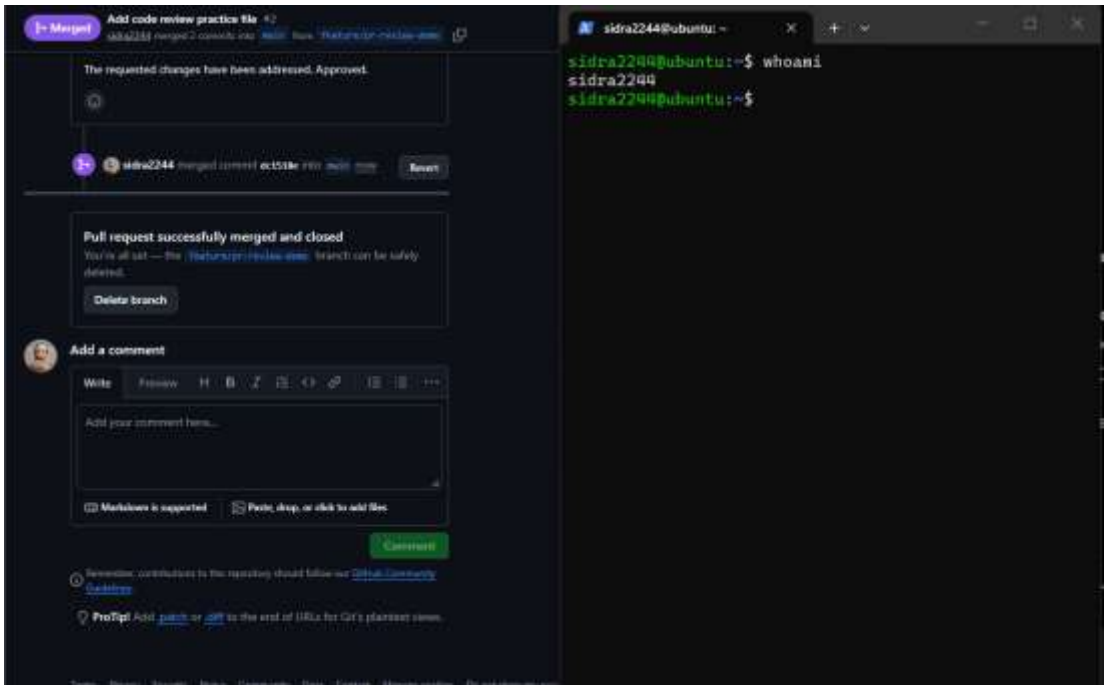


Figure 34. Pull request successfully merged

Description: Shows the final merged state of the PR.

Figure 35: Pull request branch deleted

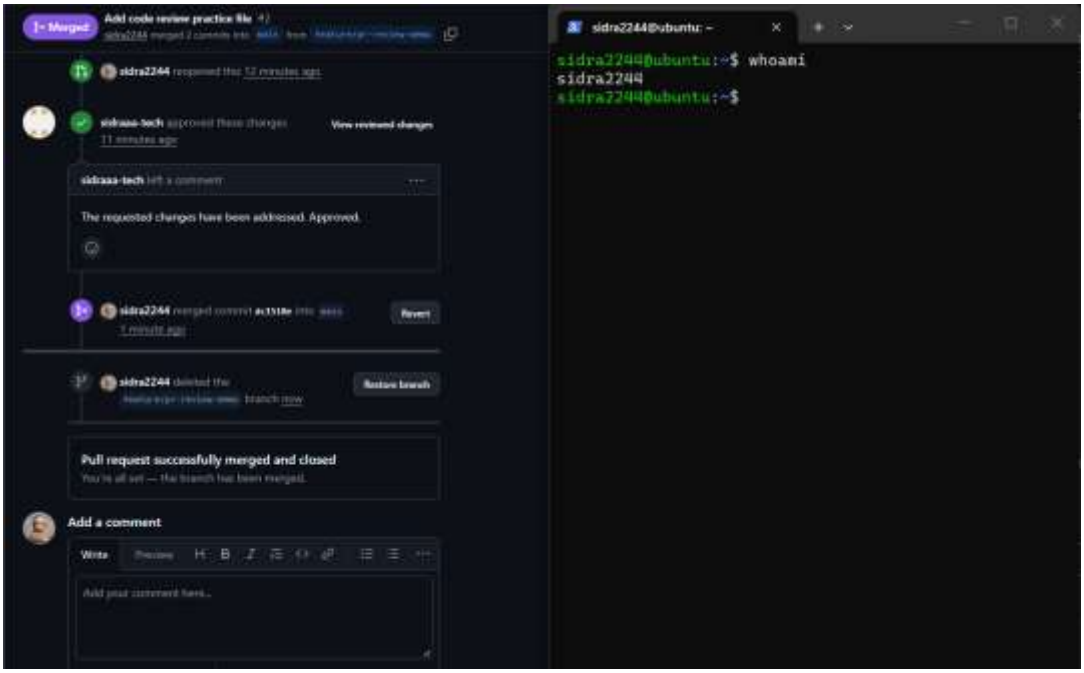


Figure 35. Pull request branch deleted

Description: Shows the branch deleted after merging.

Figure 36: Remote branch deleted in GitHub

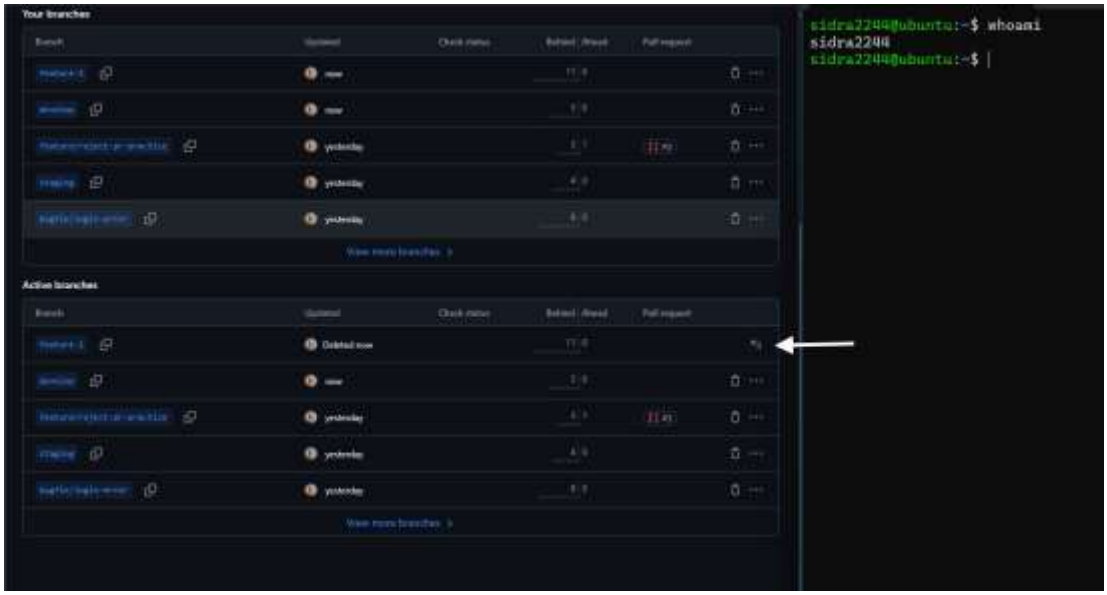


Figure 36. Remote branch deleted in GitHub

Description: Shows a completed remote branch deleted through the GitHub UI.

Figure 37: Deleting a remote branch from terminal

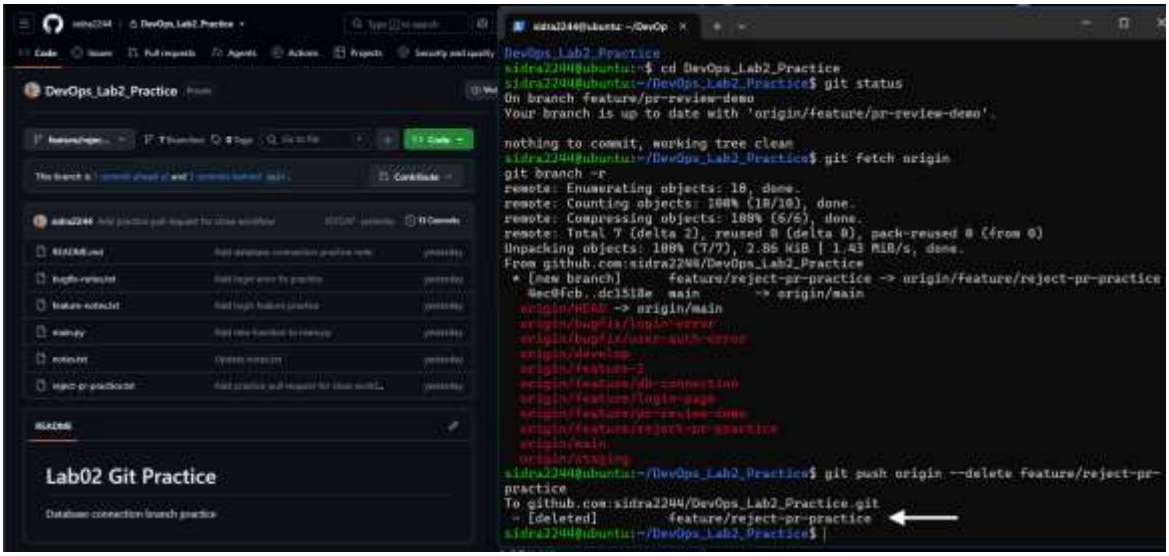


Figure 37. Deleting a remote branch from terminal

Description: Shows git push origin --delete <branch-name> and its output.

Figure 38: Collaboration commits (bonus)

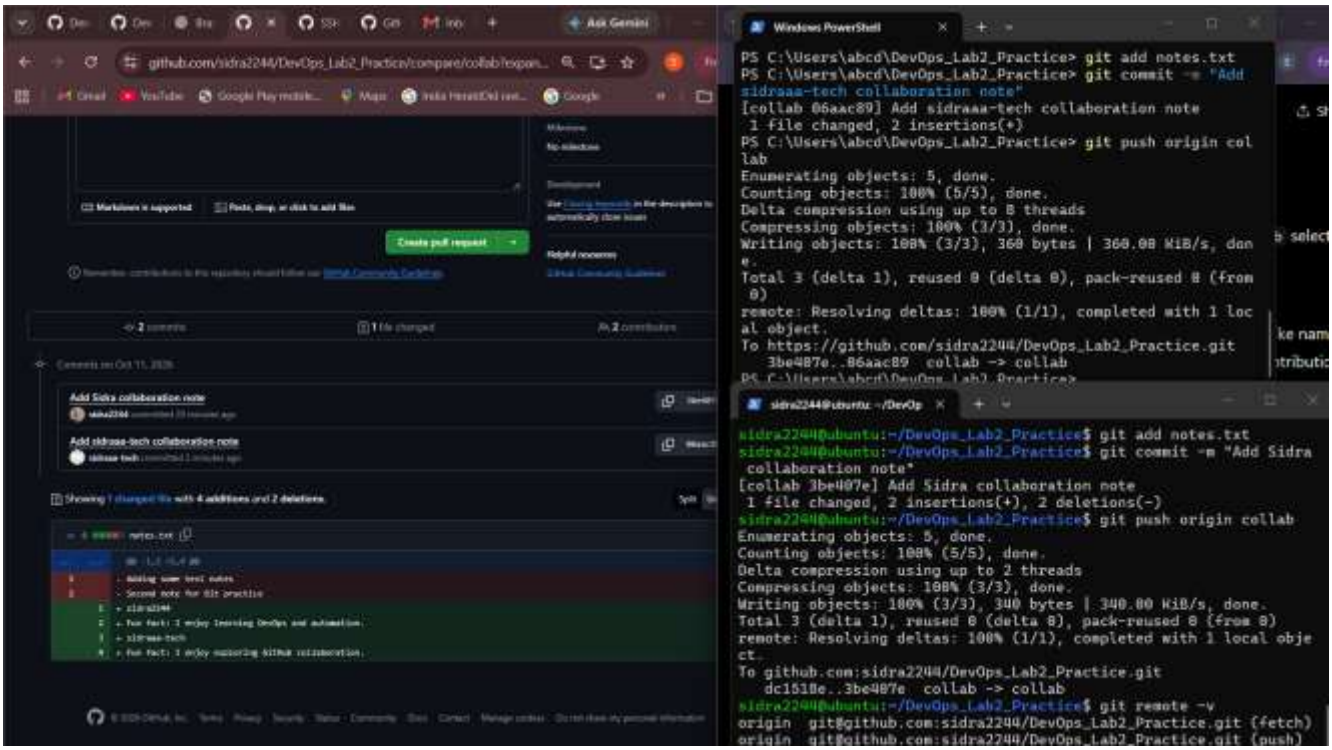


Figure 38. Collaboration commits (bonus)

Description: Shows collaborators adding their names and fun facts, then committing to collab.